

## A APPENDIX

## B DETAILS ALGORITHM OF EOFT WITH NODE AND EDGE CONSTRAINTS

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**Algorithm 2** OFT-Sinkhorn: Iterative Sinkhorn-based Algorithm for (capacited) Entropic Optimal Flow Transport with Edge and Node Constraints

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**Input:** Distance Matrix  $\mathbf{D}$ , Edge Capability  $\mathbf{S}$ , Node Capability  $\mathbf{r}$ , Marginal Difference  $\mathbf{s}$ , maximum iteration number  $L$ , Virtual-flow  $\mathbf{d}$ .

**Output:** The Allocation Matrix  $\mathbf{P}_{ij}$  for  $i, j = 1, \dots, N$

Initialize  $\mathbf{K} = \exp\left(-\frac{\mathbf{D}}{\epsilon}\right)$  element-wise.

**for**  $l = 0$  **to**  $L - 1$  **do**

    Update  $\mathbf{u}^{(l+1)}, \mathbf{v}^{(l+1)}$  according to Eq. 8

    Update  $\mathbf{q}^{(l+1)}$  according to Eq. 13

    Update  $\mathbf{G}^{(l+1)}$  according to Eq. 14

    Update  $\mathbf{K}^{(l+1)}$  by  $\exp\left(\frac{-\mathbf{D} + \mathbf{G}^{(l+1)}}{\epsilon}\right)$

**end for**

Compute  $\mathbf{P} = \text{Diag}(\mathbf{u}) \cdot \mathbf{K} \cdot \text{Diag}(\mathbf{v})$

**return**  $\mathbf{P}^* = \mathbf{P} - \mathbf{P}^\top$

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## C PROOF FOR ALGORITHM EOFT-SH, EOFT-SH-C

Here we provide the Mathematical proof for Proposition 1 and 2 in main text.

### C.1 WITHOUT CONSTRAINTS

$$\begin{aligned} \min_{\substack{\mathbf{P} \geq 0, \\ \mathbf{q} \geq 0}} \quad & \langle \mathbf{D}, \mathbf{P} \rangle - \epsilon H(\mathbf{P}) \\ \text{subject to} \quad & \mathbf{P} \mathbf{1}_N = \mathbf{q} + \mathbf{d}, \mathbf{P}^\top \mathbf{1}_N = \mathbf{q} - \mathbf{s} + \mathbf{d} \\ & \text{Diag}(\mathbf{P}) = \mathbf{d} \end{aligned} \quad (16)$$

Adding dual variables  $\mathbf{f}, \mathbf{g}, \mathbf{h}$ , the lagrange equation is then:

$$\mathcal{L} = \langle \mathbf{D}, \mathbf{P} \rangle - \epsilon H(\mathbf{P}) - \langle \mathbf{f}, \mathbf{P} \mathbf{1}_N - \mathbf{q} - \mathbf{d} \rangle - \langle \mathbf{g}, \mathbf{P}^\top \mathbf{1}_N - \mathbf{q} + \mathbf{s} - \mathbf{d} \rangle - \langle \mathbf{h}, \text{Diag}(\mathbf{P}) - \mathbf{d} \rangle \quad (17)$$

We can get partial differential of lagrange equation with respect to  $\mathbf{P}$ :

$$\frac{\partial \mathcal{L}}{\partial \mathbf{P}_{ij}} = \mathbf{D}_{ij} + \epsilon \log(\mathbf{P}_{ij}) - \mathbf{f}_i - \mathbf{g}_j = 0 \quad (i \neq j) \quad (18)$$

Then we can get :

$$\mathbf{P}_{ij} = e^{\mathbf{f}_i/\epsilon} e^{-\mathbf{D}_{ij}/\epsilon} e^{\mathbf{g}_j/\epsilon}. \quad (19)$$

Similarly:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{P}_{ii}} = \mathbf{D}_{ii} + \epsilon \log(\mathbf{P}_{ii}) - \mathbf{f}_i - \mathbf{g}_i - \mathbf{h}_i = 0 \quad (20)$$

Then we can get :

$$\mathbf{P}_{ii} = e^{\mathbf{f}_i/\epsilon} e^{-(\mathbf{D}_{ii} - \mathbf{h}_i)/\epsilon} e^{\mathbf{g}_i/\epsilon}. \quad (21)$$

Thus for the whole matrix  $\mathbf{P}$ , we can get :

$$\mathbf{P} = \text{Diag}(e^{\mathbf{f}/\epsilon}) e^{-\mathbf{D} + \text{Diag}(\mathbf{h})/\epsilon} \text{Diag}(e^{\mathbf{g}/\epsilon}). \quad (22)$$

Letting  $\mathbf{u} = e^{\mathbf{f}/\epsilon}$ ,  $\mathbf{v} = e^{\mathbf{g}/\epsilon}$ ,  $\mathbf{K} = e^{-\mathbf{D} + \text{Diag}(\mathbf{h})/\epsilon}$ , we can get the optimal solution from  $\mathbf{P} = \text{Diag}(\mathbf{u}) \mathbf{K} \text{Diag}(\mathbf{v})$ .